

Clinical Natural Language Technology for Health Care: Past, Present, and Future Approaches

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The Concept

Clinical natural language technology is the set of methods that turn unstructured health care information — clinician notes, faxed records, scanned forms, imaging reports — into a form software can act on. It spans several distinct but converging technologies: natural language processing (NLP) for parsing text, optical character recognition (OCR) for digitizing scanned or faxed documents, computer vision (CV) for interpreting images and imaging reports, large language models (LLMs) for extracting and summarizing meaning from free text, and large multimodal models (LMMs) that reason jointly across text and images. For a healthcare informatics company processing high volumes of unstructured clinical, billing, and correspondence content across treatment, payment, and operations (TPO) use cases, the maturity of these technologies directly determines how much of that content can be structured, verified, and acted on without manual review. Cotiviti's payment-integrity work in particular depends on reading exactly this kind of unstructured material — medical records, payer-provider contracts, clinical documentation attached to a claim — so the pace of progress in clinical NLP is not an academic curiosity for the company; it is a direct input to how much of that review work can be automated versus staffed.

Trends: Past, Present, and Future

Past. Before deep learning, clinical NLP relied on rule-based pattern matching, curated medical dictionaries, and statistical models such as conditional random fields trained for narrow extraction tasks — each new use case required its own hand-built pipeline (Wu et al., 2020). Digitizing scanned or handwritten records was a separate, brittle problem: even combining multiple general-purpose OCR engines produced meaningful error rates on real clinical form fields (Rasmussen et al., 2012), a constraint this assessment's proof of concept reproduces directly.

Present. Transformer-based language models changed both halves of the problem. Domain-pretrained models such as BioBERT and ClinicalBERT closed much of the gap on biomedical entity recognition and note-level prediction tasks (Lee et al., 2020; Huang et al., 2019). More recently, general-purpose LLMs have absorbed enough medical knowledge to exceed passing-level performance on medical licensing exam benchmarks (Singhal et al., 2023) and are already deployed at the point of care: a 2026 multi-site study of ambient AI documentation scribes found meaningful, if inconsistent, time savings for clinicians (Aguilar, 2026). The practical shift for a company like Cotiviti is that a single LLM call with a structured output schema can now do what used to require a purpose-built extraction pipeline per document type (Thirunavukarasu et al., 2023) — the same pattern this assessment's proof of concept demonstrates on a synthetic clinical note.

Future. The next step is multimodal: models such as Med-PaLM M reason jointly over clinical text, medical imaging, and genomic data with a single set of weights rather than separate pipelines stitched together after the fact (Tu et al., 2024). For TPO workflows that already combine text (notes, policies) with image attachments (itemized bills, imaging studies), this points toward a single review pipeline instead of parallel OCR, NLP, and CV systems maintained independently.

Opportunities and Threats

Opportunities. LLM-based extraction is markedly more resilient to the noisy, inconsistent text that OCR produces from real scanned documents than older rule-based pipelines were, and it requires far less engineering investment per new document type or use case — no bespoke named-entity model to train and maintain for each one. Deployed carefully, this can reduce the manual abstraction burden on clinical and payment-integrity reviewers, and, per Aguilar (2026), measurably reduce documentation time even in today's imperfect deployments.

Threats. The same generality that makes LLMs flexible also makes them unreliable in high-stakes ways: a 2025 adversarial study found leading LLMs repeated a single fabricated clinical detail in 50–83% of test cases when it was planted in a prompt (Omar et al., 2025). Regulatory exposure is real and growing — the FDA's AI/ML-based Software as a Medical Device action plan sets expectations for change-control and real-world monitoring (U.S.

Food and Drug Administration, 2021), and HHS's 2024 proposed HIPAA Security Rule updates would explicitly extend compliance obligations to AI systems and training data that touch electronic protected health information (U.S. Department of Health and Human Services, 2024). Bias is not hypothetical either: a widely used population-health algorithm was shown to under-flag Black patients for extra care because it used cost, not need, as its training proxy — correcting it nearly tripled the share of Black patients identified as high-risk (Obermeyer et al., 2019). Any TPO-facing model trained on historical cost or utilization data inherits this exact risk.

Recommended Options for Cotiviti

1. **Pilot a governed extraction service.** Apply the OCR-plus-LLM pattern demonstrated in the accompanying proof of concept to a real, high-volume unstructured input Cotiviti already receives — for example, faxed medical records or provider correspondence attached to payment-integrity claims — with a human reviewer confirming any output before it reaches an automated decision. Scope the pilot narrowly at first (one document type, one downstream workflow) so accuracy and time-savings can be measured against the existing manual process before any expansion decision is made.
2. **Build the AI governance layer first, not after.** Before scaling any LLM-based TPO tool, invest in the AI Governance Program already referenced in Cotiviti's own intern job description: hallucination spot-checks against source documents, bias audits on any model trained with cost or utilization signals (informed directly by the Obermeyer et al. finding above), and a change-control process aligned to the FDA's AI/ML SaMD framework and HHS's proposed ePHI rules. Treat this as a standing function, not a one-time review, since both the underlying models and the regulatory expectations around them are still changing quickly.
3. **Track multimodal models as a medium-term bet.** Multimodal systems like Med-PaLM M are not yet a build priority, but Cotiviti should monitor them as a path toward unifying text- and image-based TPO review into a single pipeline over the next two to three years, revisiting the build decision once a narrow-scope model is available for evaluation rather than committing engineering time to it today.

These three options are sequenced deliberately: piloting first, on a narrow and easily reviewed scope, produces the evidence needed to justify investment in governance; governance, in turn, is what makes it defensible to widen the pilot's scope later without repeating the same hallucination, privacy, and bias risks at larger scale. The accompanying proof of concept is a small, concrete illustration of the first step — an OCR-plus-LLM pipeline that turns a scanned clinical note into structured fields in seconds rather than requiring a purpose-built extraction model — not a finished product, but evidence that the underlying pattern is fast to stand up and worth piloting properly.

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